



OrganaDx acts as a preliminary tester before food products go to testing labs (Credit: www.mydxlife.com)

certain pesticides, weedicides, and fertilisers in the farming methods as well as the use of irradiation, industrial solvents, or synthetic food additives.

The labelling of organic food products is the need of the hour to strengthen consumers' trust. Organic products often have a number of labels on them depending on the place of their origin and testing, but the USDA (United States Department of Agriculture) organic label is the only one that's federally certified by accredited agents. Here are some of the general standards fixed for the production and processing of food products, which can then be labelled or certified as organic:

- Food products are grown and processed according to the guidelines based on several factors such as soil quality, animal raising practices, pest and weed control, and use of additives
- Producers rely on natural substances and physical, mechanical, or biologically based farming methods
- Product is assigned as organic if it is grown on soil that has used no prohibited substances (pesticides, weedicides, and fertilisers) for three years prior to harvest
- Similarly, regarding organic meat, the regulations require that animals are raised in living conditions accommodating their natural behaviour, fed hundred per cent organic feed and forage, and not administered antibiotics or hormones
- Regarding processed multi-

ingredient foods, the organic standards specify additional regulations prohibiting artificial preservatives, colours, or flavours and require that their ingredients are organic

- No organic food is grown or handled using genetically modified organisms

A comparison between organic and conventional food products

With only some recognised differences between nutrient and anti-nutrient contents of organically and conventionally produced food, the variable nature of food production, shipping, storage, and handling it is difficult to generalise the results. Moreover, from the perspective of science and consumers, there is not enough evidence to support the claims that eating organic food is safer or healthier in comparison to conventional food. Even the several claims suggesting that organic food tastes better are not supported by tests. Further, organic farming has higher production costs and lower yields, and thus higher consumer prices as compared to conventional food.

But still, the demand for organic food is high as consumers are concerned about their personal health and environment. Becoming familiar with the organic label and understanding its claim empower consumers to make informed decisions about the food they purchase. Consumers can look for the identity of the certifier on

a packaged product for verifying that the organic product meets organic standards. However, some people are always trying to sell conventional food as organic to earn more money.

Authentication methods

The consumer's interest in organic foods is increasing, and so is the need for robust analytical tools for their authenticate labelling and certification as organic. The certification of food products as 'organic' is protected by laws in every country and can be used by producers who have received official certification according to standards. In some countries, organic food production is being developed as a self-regulated industry. The food products marketed as organic are to be produced in a way that complies with organic standards set by national governments and international organic industry trade organisations.

On these lines, currently, the European Union, the United States, Canada, Japan, and many other countries require producers to obtain special certification based on government-defined standards to market food as organic. Products are tested for authenticity before being certified as organic. This certification acts as a guarantee to consumers that the organic food items have been produced and processed without the use of pesticides and chemical fertilisers and do not contain any contaminant.

There are different biomarkers/analytical approaches available for the authentication of organic produce. Services of organic food testing laboratories are often utilised to analyse organic foods for their nutrition content and for the added health benefits they might claim. Food testing assures consumers that the products they are purchasing are authentically organic.

Food Safety and Standards Authority of India (FSSAI) has recently operationalised the Food Safety and Standards (Organic Foods) Regulation, 2017 so that the food standards and safety are maintained. Food testing laboratories provide these services to ensure that organic food productions, as well as processed products, are free of chemical and microbial contaminants. Before certifying food items, the certifying authorities like